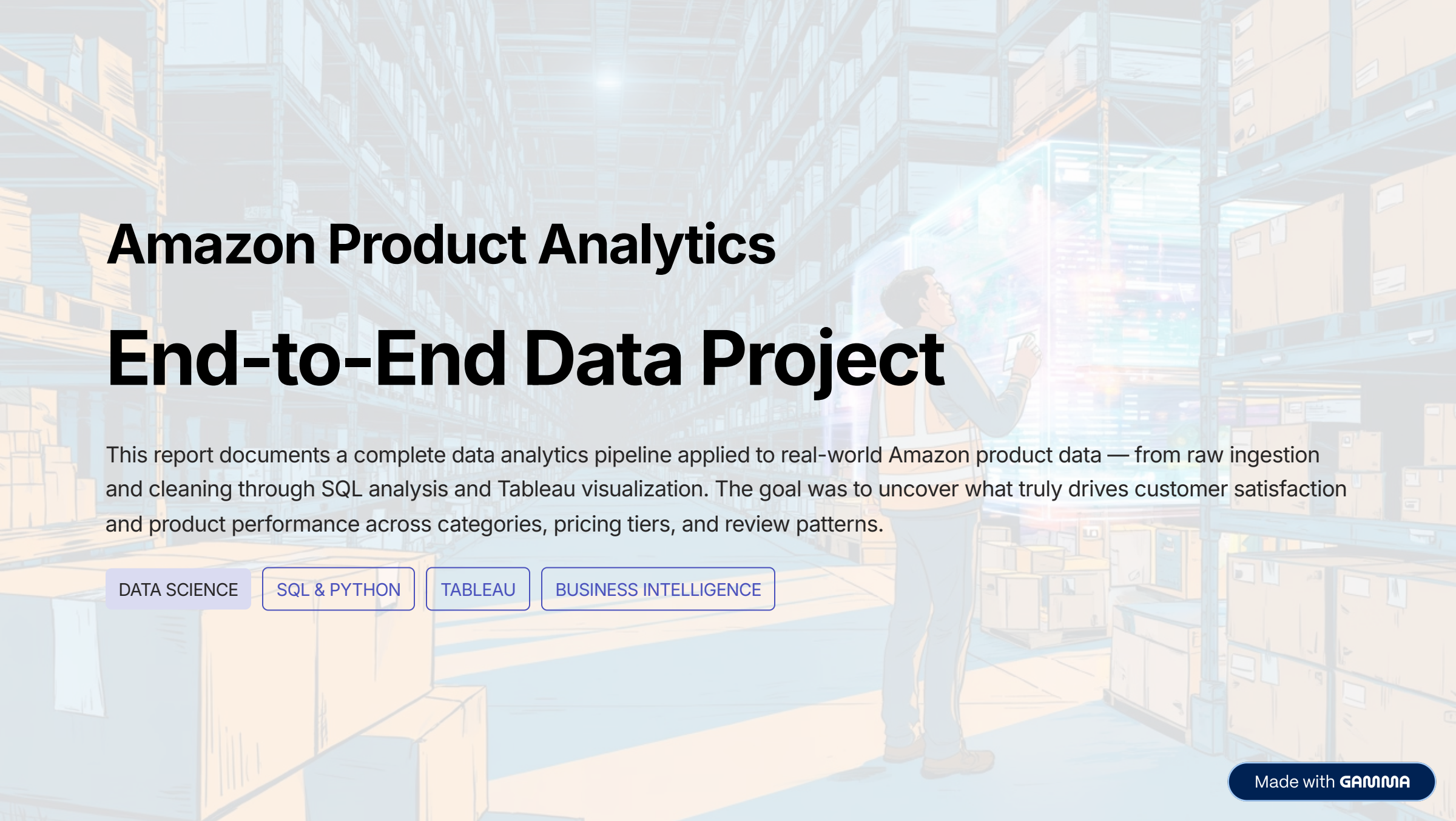




Amazon Product Analytics End-to-End Data Project

This report documents a complete data analytics pipeline applied to real-world Amazon product data — from raw ingestion and cleaning through SQL analysis and Tableau visualization. The goal was to uncover what truly drives customer satisfaction and product performance across categories, pricing tiers, and review patterns.

DATA SCIENCE

SQL & PYTHON

TABLEAU

BUSINESS INTELLIGENCE

Problem Statement

Amazon's marketplace generates an enormous volume of product and review data every day. While this data is rich with potential insights, it rarely speaks for itself — it requires careful extraction, transformation, and analysis to reveal meaningful patterns. This project was designed to answer a core set of business questions that matter to product managers, category leads, and pricing strategists operating within or alongside the Amazon ecosystem.

Specifically, the analysis focuses on three interconnected dimensions of product performance. First, **pricing impact**: do discounts and price positioning meaningfully influence how customers rate products? Second, **customer satisfaction**: what patterns emerge in review behavior, and how do ratings correlate with product attributes? Third, **category performance**: do certain product categories consistently outperform others in terms of customer sentiment, and what does that tell us about buyer expectations?

These questions are not merely academic. For sellers and brands on Amazon, understanding the relationship between price, category, and satisfaction can inform listing strategy, promotional planning, and inventory decisions. For internal Amazon teams, these patterns can surface opportunities to improve the customer experience, refine recommendation algorithms, or identify categories that may need additional quality controls. The analysis also touches on customer behavior signals — such as review length and rating distribution — that can serve as proxies for engagement depth and purchase confidence.

The challenge is compounded by the nature of real-world e-commerce data. Product listings include mixed data types, inconsistent formatting (currency symbols, varied rating scales), and missing values that must be resolved before any analysis can begin. Reviews introduce additional complexity, as text fields vary widely in length and sentiment expression. A rigorous data preparation phase was therefore essential before any modeling or visualization could take place. The sections that follow walk through each stage of this pipeline in detail.

Pricing Impact

Does discounting meaningfully influence customer ratings and purchase sentiment?

Customer Satisfaction

What patterns emerge in review behavior and rating distributions across products?

Category Performance

Which product categories consistently outperform others in customer sentiment?

Data Overview

The dataset used in this project is a structured Amazon product table containing 16 columns that span product metadata, pricing information, and customer review data. It represents a realistic cross-section of e-commerce product listings, with enough breadth to support multi-dimensional analysis across pricing, ratings, categories, and textual review attributes. The data was sourced in a raw format and required significant preprocessing before it could be used for analytical queries and visualizations.

Dataset Structure

The 16 columns cover several domains of information. Product identity fields include product name, ASIN or SKU equivalent, and category classification. Pricing fields capture list price, actual selling price, and discount information — though in raw form these included currency symbols and inconsistent formatting that required normalization. Review-related fields include star rating, review count, and review text length, which served as a proxy for review depth and customer engagement.

The dataset includes multiple data types — strings for product names and categories, floats and integers for pricing and ratings, and datetime fields for review timestamps. This heterogeneity is typical of real-world e-commerce exports and was a primary driver of the data cleaning effort described in a later section. Missing values appeared across several columns, particularly in pricing fields for products without active discounts and in review text fields for listings with limited customer feedback.

Before any analysis could begin, the data required a thorough quality assessment. This included checking for duplicate records, validating rating ranges (ensuring all values fell within the expected 1–5 star scale), and confirming that category labels were consistent and free of typos or alternate naming conventions. The cleaning and standardization work laid the foundation for reliable SQL queries and accurate Tableau visualizations downstream.

Key Data Attributes

16 Columns

Spanning product identity, pricing, reviews, and category metadata

Multiple Data Types

Strings, floats, integers, and datetimes requiring type normalization

Missing Values

Present across pricing and review fields, requiring imputation or exclusion

Tools & Technologies

This project leveraged a modern data analytics stack, with each tool serving a distinct role in the pipeline. The selection reflects industry-standard practices for end-to-end data projects — from data wrangling and database querying to visualization and reporting. Together, these tools enabled a seamless workflow from raw data to actionable insight.



Python & Pandas

The primary engine for data ingestion, cleaning, and transformation. Pandas DataFrames enabled efficient manipulation of the 16-column dataset, including type conversions, missing value handling, and feature creation. Python scripts were used to automate repetitive cleaning tasks and ensure reproducibility across analysis iterations.



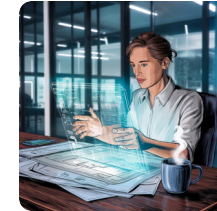
PostgreSQL & SQL

Used for structured analytical querying once the cleaned data was loaded into a relational database. SQL enabled powerful aggregations, window functions, and joins to answer the core business questions around category performance, discount impact, and customer behavior patterns at scale.



Tableau

The visualization layer of the project. Tableau transformed SQL query results into interactive dashboards that communicate findings to business stakeholders. Charts, filters, and calculated fields were used to make the insights accessible and explorable without requiring technical expertise from the end user.

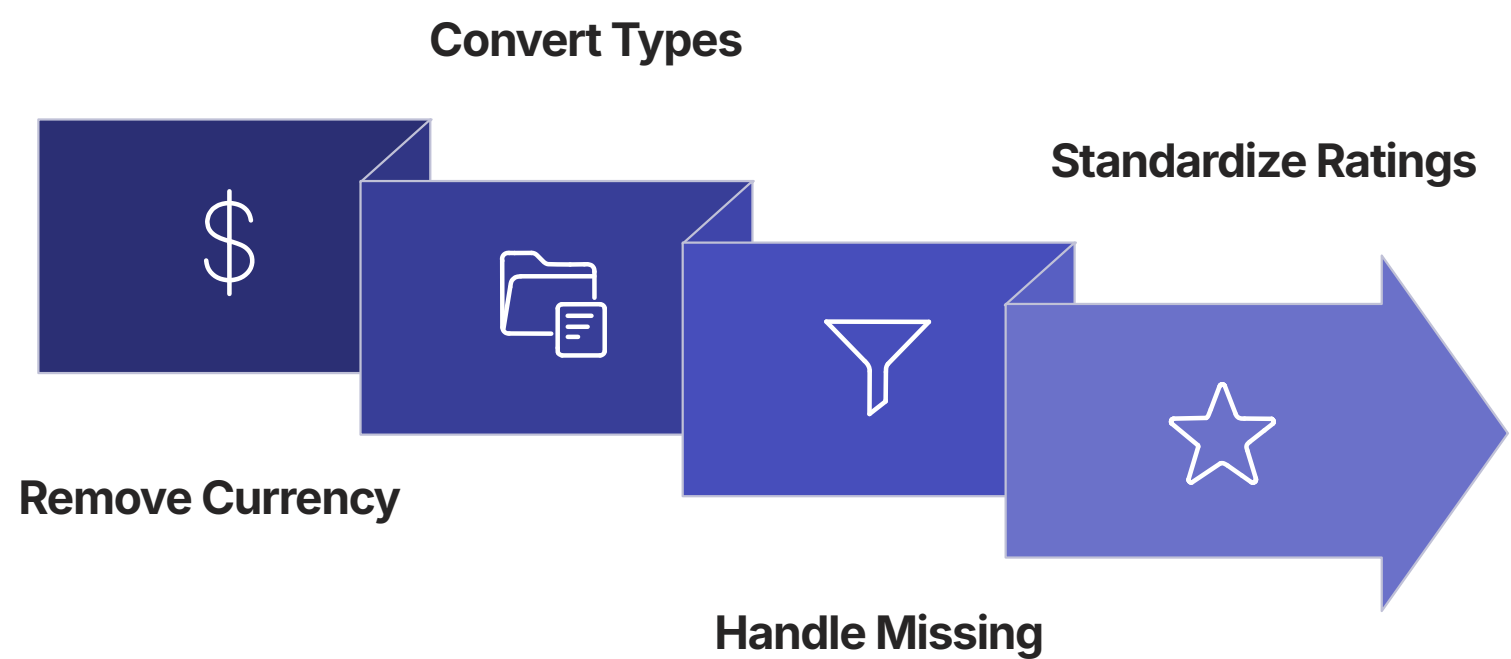


Gamma AI

Used to structure and present this report. Gamma AI assisted in organizing the analytical narrative, generating visual layouts, and producing a polished final document suitable for stakeholder review and presentation delivery.

Data Cleaning Process

Raw e-commerce data is rarely analysis-ready. The Amazon product dataset required a systematic cleaning pipeline to resolve formatting inconsistencies, type mismatches, and data quality issues before any meaningful analysis could be performed. This phase is often the most time-consuming part of a data project, and in this case it accounted for a significant portion of the total effort — but it was essential groundwork that ensured all downstream results were trustworthy.



The first step addressed a common issue in scraped or exported e-commerce data: currency symbols embedded in price fields. Values like "\$19.99" cannot be used in arithmetic operations until the dollar sign is stripped and the string is cast to a float. This was handled using pandas string methods combined with type casting, ensuring all pricing columns were numeric and ready for aggregation.

Data type conversion extended beyond pricing fields. Review timestamps needed to be parsed into proper datetime objects to enable time-based filtering and trend analysis. Product IDs and category fields were validated as strings and checked for unexpected whitespace or encoding artifacts that could cause join failures in SQL. Each column was inspected individually, and a data dictionary was maintained throughout to track the original and cleaned state of every field.

Missing values presented a more nuanced challenge. Some products had no discount information because they were never on sale — in these cases, a null was semantically correct and was preserved rather than imputed. Other missing values, such as absent review counts or ratings, were investigated to determine whether they represented data collection gaps or genuinely unrated products. Records with critical missing fields were excluded from analyses that depended on those fields, while less critical gaps were filled using median imputation or flag variables.

Finally, rating standardization ensured that all star ratings fell within the expected 1–5 range. A small number of outlier values were identified and either corrected (where the source was clearly a formatting error) or flagged for exclusion. This step was critical because rating distributions form the backbone of the satisfaction analysis, and any skew from uncorrected outliers would have compromised the validity of subsequent findings.

Feature Engineering

Once the raw data was cleaned, the next phase was feature engineering — creating new derived variables that capture dimensions of product performance not directly represented in the original columns. These engineered features enriched the analytical model and enabled more nuanced SQL queries and Tableau visualizations. Three key features were developed for this project, each designed to surface a different aspect of product or customer behavior.



discount_value

Calculated as the difference between list price and actual selling price, expressed both as an absolute dollar amount and as a percentage. This feature quantifies the depth of discounting for each product, enabling direct comparison between heavily discounted items and those sold at or near list price. It serves as the primary variable for testing whether price reductions correlate with higher customer ratings.



review_length

Derived by counting the number of words or characters in each review text field. Review length acts as a proxy for engagement depth — longer reviews tend to reflect more thoughtful customer experiences, while very short reviews may indicate impulsive ratings or low engagement. This feature was used to segment customers by review behavior and test whether review depth correlates with rating extremity.



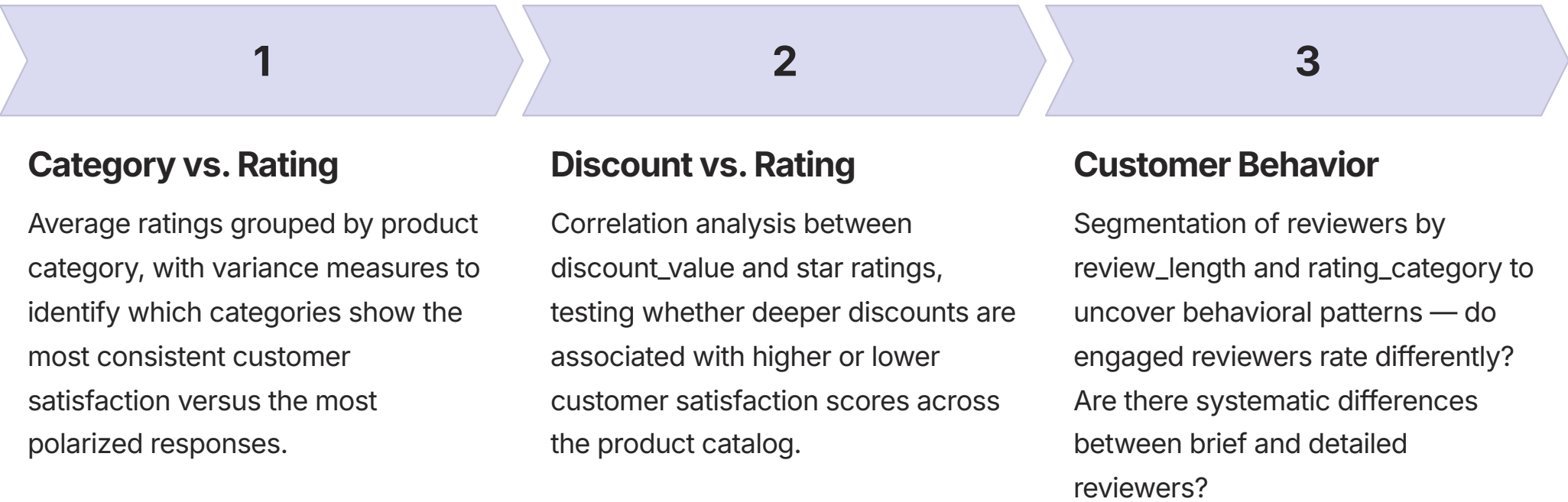
rating_category

A binned categorical variable derived from the continuous star rating field, grouping ratings into tiers such as "Poor" (1–2), "Average" (3), and "Excellent" (4–5). This transformation simplifies rating distributions for cross-category comparisons and enables chi-square and frequency analyses that would be less intuitive with raw numeric ratings. It also aligns with how business stakeholders typically think about satisfaction tiers.

These three features were added to the cleaned dataset before loading into PostgreSQL, ensuring they were available for SQL-based aggregation and filtering. In Tableau, they served as dimensions and measures for calculated fields, enabling dynamic segmentation of products by discount depth, review engagement, and satisfaction tier. The feature engineering phase demonstrated how relatively simple transformations can significantly expand the analytical surface area of a dataset without requiring additional data sources.

SQL Analysis

With the cleaned and enriched dataset loaded into PostgreSQL, the analytical core of the project began. SQL queries were structured to address each of the three primary research questions, using aggregations, subqueries, and window functions to extract patterns from the data. The queries were designed to be both analytically rigorous and interpretable by non-technical stakeholders when translated into visual form.



The category vs. rating analysis revealed meaningful variation across product types. Some categories consistently clustered around 4.0–4.5 average stars, while others showed wider dispersion, suggesting that customer expectations differ substantially by category. This finding has direct implications for how sellers should benchmark their products — a 3.8-star rating may be strong in one category and weak in another.

The discount vs. rating query was perhaps the most counterintuitive. Across the dataset, the correlation between discount depth and star rating was surprisingly weak. Products with deep discounts did not consistently receive higher ratings than those sold at full price, suggesting that price reductions alone do not drive satisfaction. This insight challenges a common assumption in promotional strategy and points toward product quality as a more dominant factor.

Customer behavior analysis added another layer of nuance. Reviewers who wrote longer reviews tended to give more moderate ratings, clustering around 3–4 stars, while shorter reviews showed greater extremity — more 1-star and 5-star ratings. This pattern suggests that thoughtful, detailed feedback may be more calibrated, while impulsive reviews skew toward the extremes. These behavioral segments can inform how brands interpret and respond to their review profiles.

Key Insights

The analytical work surfaced several findings that have direct implications for product strategy, pricing decisions, and customer experience management on Amazon. These insights are grounded in the data but framed in terms that are actionable for business stakeholders who may not be familiar with the technical details of the analysis pipeline.

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Core Findings

Across pricing, category, and behavior dimensions

Discounts Don't Drive Ratings

One of the most significant findings is that discount depth has a surprisingly weak relationship with customer satisfaction scores. Products offered at steep discounts did not consistently receive higher ratings than those sold at full price. This suggests that customers evaluate products primarily on their inherent quality and fit for purpose, rather than on the perceived value of a deal. For pricing strategists, this implies that promotional discounts should be viewed as a volume or acquisition tool rather than a satisfaction driver.

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Data Columns

Analyzed across product, pricing, and review dimensions

Category Is a Stronger Satisfaction Driver

Product category emerged as a much stronger predictor of rating patterns than pricing variables. Certain categories consistently showed higher average ratings and lower variance, while others exhibited more polarized customer responses. This finding suggests that customer expectations are heavily shaped by category norms — buyers of electronics may have different satisfaction thresholds than buyers of home goods. Brands should benchmark their ratings against category-specific baselines rather than platform-wide averages.

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Star Rating Scale

Standardized and analyzed for cross-category comparison

High Variability in Product Performance

Even within the same category, individual products showed substantial variation in ratings and review patterns. This variability indicates that product-level factors — such as quality control, listing accuracy, and fulfillment experience — play a critical role in shaping customer sentiment. It also suggests that aggregate category-level metrics can mask important product-level opportunities and risks. Brands should monitor individual SKU performance rather than relying solely on category averages.

Tableau Dashboard

The analytical findings were translated into an interactive Tableau dashboard designed for business stakeholders. The dashboard organizes the key dimensions of the analysis — category performance, discount impact, and customer behavior — into a cohesive visual interface that allows users to explore the data without requiring SQL or Python expertise. Filters and drill-down capabilities enable stakeholders to investigate specific categories, price ranges, or rating tiers on their own.



The first dashboard view focuses on category-level performance, displaying average ratings by product category alongside distribution metrics that highlight variance. This view makes it immediately clear which categories are outperforming or underperforming the platform average, and it allows stakeholders to drill into specific categories to examine individual product contributions. Color encoding and sorted bar charts enable quick identification of top and bottom performers.

The second view examines the relationship between discount depth and customer ratings, visualizing the weak correlation identified in the SQL analysis. Scatter plots and binned comparison charts show that products across the full discount spectrum receive similar rating distributions, reinforcing the conclusion that price reductions alone do not drive satisfaction. This view is particularly useful for pricing teams evaluating the ROI of promotional strategies.

Both dashboard views include interactive filters for category, price range, and rating tier, enabling stakeholders to explore the data from multiple angles. The dashboard was designed with a non-technical audience in mind — labels are clear, color choices follow accessibility best practices, and tooltips provide context without overwhelming the viewer. The goal was to make the analytical findings accessible and actionable for product managers, category leads, and business analysts who need to make decisions based on this data.

Conclusion & Recommendations

Product quality matters more than pricing strategy in determining customer satisfaction on Amazon's marketplace.

This project demonstrates the power of a structured, end-to-end analytics approach to answering real business questions. By moving systematically from raw data through cleaning, feature engineering, SQL analysis, and visualization, the project produced findings that are both analytically sound and practically relevant. The central conclusion — that product quality, not discount depth, is the primary driver of customer satisfaction — has meaningful implications for how brands and sellers should allocate their resources on the Amazon platform.

For product managers and category leads, the findings suggest that investment in product quality, accurate listings, and reliable fulfillment will yield stronger satisfaction outcomes than aggressive promotional pricing. Discounts may drive short-term volume, but they do not appear to improve the customer experience in a lasting way. Brands should treat pricing promotions as a tactical tool rather than a strategic lever for building long-term reputation.

Invest in Quality
Prioritize product quality and listing accuracy over discount-driven strategies to build sustained customer satisfaction and brand reputation.

Benchmark by Category
Use category-specific rating baselines rather than platform-wide averages when evaluating product performance and setting improvement targets.

Monitor Individual SKUs
Track product-level metrics closely, as high variability within categories means aggregate data can mask critical opportunities and risks.

Extend the Analysis
Future work should incorporate sentiment analysis on review text, time-series trend analysis, and competitive benchmarking to deepen these findings.

Looking ahead, several avenues for extending this work present themselves. Incorporating natural language processing on review text could surface sentiment patterns that star ratings alone do not capture. Time-series analysis could reveal whether satisfaction trends are improving or deteriorating for specific categories or brands. And integrating competitive data — such as competitor pricing and ratings — would enable a more complete picture of market positioning. The pipeline built for this project provides a solid foundation for all of these extensions.

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This project showcases a complete analytics workflow: from raw Amazon product data to cleaned datasets, engineered features, SQL queries, and interactive Tableau dashboards — delivering actionable business intelligence at every step.